

Platonic G: Gravitation from Geometry

Boris V. Kochenov
kochenovforwork@gmail.com

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Abstract

This note presents an empirical geometric observation: the gravitational constant G can be expressed purely geometrically using only the golden ratio ϕ , the integer 31, and $\sqrt{5}$. No CODATA values are used in the derivation.

1 Main Result

The gravitational constant admits a geometric representation:

$$G \times 10^{11} = \frac{31^2 \times \sqrt{5}}{\phi^{12}} = 6.67354674496161812375 \dots$$

Source	Value ($\times 10^{-11}$)	Gap from G_K
Schlamminger (2014)	6.6729 ± 0.0005	+0.010%
G_K (this work)	$6.67354674496 \dots$	—
CODATA 2018	6.67430 ± 0.00015	−0.011%
Labs disagree	0.021%	

G_K lies between both measurements. Each gap is smaller than the mutual disagreement between laboratories. Inputs: ϕ , 31, $\sqrt{5}$ only. No measured constants. No fitting.

2 The Generator Identity

The expression $\phi^{12}/\sqrt{5}$ is the Binet approximation to $F_{12} = 144$, the unique perfect square in the Fibonacci sequence. Direct evaluation yields:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8})$$

where $720 = 6!$ — a unique intersection of the exponential sequence ϕ^n , the factorial sequence $m!$, and Fibonacci numbers F_n at indices $n = 12$, $m = 6$. No comparable alignment exists for nearby indices. The absolute deviation is $|\Delta| = 1.39 \times 10^{-8}$, with relative precision $|\Delta|/144 = 9.7 \times 10^{-11}$.

3 The Residual $1/720$ — Three Independent Domains

The same residual $1/720$ appears independently in three physical domains:

Domain	Formula
Geometry	$\phi^{12}/\sqrt{5} - 144 = 1/720$
Nuclear	$m_n = m_p \times (1 + 1/720)$
EM	$\alpha^{-1} = 144 - 7 - 1/720$

Three independent physical domains. One residual. The geometry identity is exact to 10^{-8} . The nuclear result (neutron-to-proton mass ratio) is verified against CODATA to 0.001%. The electromagnetic result uses the rounded value $\alpha^{-1} = 137$.

4 Open Question

The integer $31 = 2^5 - 1$ (Mersenne prime) is the structural anchor of the formula. Its geometric origin remains under investigation.

5 What This Is Not

This is an empirical observation. No dynamical model is claimed. No CODATA values were used in the derivation. The formula is presented as a geometric identity whose physical significance, if any, remains to be established.

6 Full Precision Reference

$$G_K = 6.6735467449616181237591997717904236079132174548715 \\ 1565557657672182332559409890102447913340675769273 \times 10^{-11}$$

Derived from: $G_K = 31^2 \times \sqrt{5}/(\phi^{12} \times 10^{11})$.

Geometric Origin of 10^{11} in the Gravitational Constant

Abstract

This note presents an empirical geometric observation: the number 10^{11} that appears in the gravitational constant G is not a decimal artifact. It arises naturally from four turns of a 3D golden spiral (similar to a Fibonacci spiral but rising in height) and the addition of a micro-step $1/720$.

Basic spiral block

$$S = \phi^6 \times 31 \approx 556.272429, \phi = (1 + \sqrt{5})/2.$$

Spiral inclination

$$\kappa = \log_{10}(S) \approx 2.745288.$$

Four turns without micro-step

$$4\kappa \approx 10.981150.$$

Target height

$$\log_{10}(10^{11}) = 11.000000.$$

Difference

$$11.000000 - 10.981150 = 0.018850 \text{ } (\approx 0.171\% \text{ of } 11).$$

Micro-step

One full turn consists of 12 micro-steps, each $1/720$:

$$\frac{12}{720} = \frac{1}{60} \approx 0.016667$$

covers the gap almost exactly.

Residual

$$0.018850 - 0.016667 = 0.002183 \text{ } (\approx 0.020\% \text{ of } 11), \text{ fractal noise.}$$

Resonant relation

$$\log_{10}(10^{11}) \approx 4\log_{10}(\phi^6 \times 31) + \frac{12}{720}, \quad 10.981150 + 0.016667 = 10.997817.$$

Equivalent forms of G_{Koch}

$$G_{\text{Koch}} = \frac{31^2 \sqrt{5}}{\phi^{12} \times 10^{11}} = \frac{31^2}{(144 + 1/720) \times 10^{11}} \approx 6.67354674496 \times 10^{-11}.$$

Conclusion

The Universe is not a set of arbitrary constants. It is one spiral, read at different scales.

A High-Precision Identity Linking $\phi^{12}/\sqrt{5}$, the Fibonacci Square F_{12} , and a Micro-Deviation of $1/720$

Abstract

This note reports a high-precision numerical identity relating $\phi^{12}/\sqrt{5}$ to the Fibonacci number $F_{12} = 144$ with a correction term of magnitude $1/720$. The deviation is of order 10^{-8} , significantly below typical experimental uncertainties in physics.

Statement of the Identity

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad X = \frac{\phi^{12}}{\sqrt{5}} = 144.001388875432 \dots$$

$$144 + \frac{1}{720} = 144.001388888888 \dots$$

$$\Delta = X - \left(144 + \frac{1}{720}\right) = -1.394 \times 10^{-8}$$

Thus,

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8}).$$

Relation to Fibonacci Numbers

Using Binet's formula: $F_{12} = 144$. Hence,

$$\frac{\phi^{12}}{\sqrt{5}} = F_{12} + \varepsilon, \quad \varepsilon = 0.001388875432 \dots \approx \frac{1}{720}.$$

Relative deviation $\approx 9.68 \times 10^{-11}$.

Precision Analysis

Absolute deviation $|\Delta| \approx 1.39 \times 10^{-8}$, relative deviation $|\Delta|/X \approx 9.68 \times 10^{-11}$.

Remarks

The rational correction $1/720$ is not predicted by standard Fibonacci identities.

Conclusion

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} \pm 1.4 \times 10^{-8}.$$

The Integer 31 as a Discrete Geometric Node in Kochenov G

Abstract

The integer 31 functions as a discrete stabilizing node in the Kochenov gravitational formula $G = 31^2 \sqrt{5} / (\phi^{12} \times 10^{11})$. It emerges as the optimal integer satisfying quadratic closure $31^2 = 961$ (gap 0.001%) and scaling optimum $(\phi^6 \times n)^4 = 10^{11}$ giving $n \approx 31.338$. No neighbouring integer (30,32) satisfies both constraints.

Spiral Block

$S = \phi^6 \times 31 = 556.272429\dots$, $\kappa = \log_{10}(S) = 2.745288$, $4\kappa = 10.981150$, gap closed by $12/720 = 0.016667$.

Two Constraints

Quadratic closure: $31^2 = 961 = G_{\text{Koch}} \times 10^{11} \times 144$. Scaling optimum: $(\phi^6 \times n)^4 = 10^{11} \rightarrow n \approx 31.338$.

Neighbouring Integers

n	G_n gap from CODATA	S^4 gap from 10^{11}	Note
30	6.358%	16.018%	too small
31	0.011%	4.247%	optimal discrete node
32	6.544%	8.718%	too large
31.338	$\sim 0\%$	$\sim 0\%$	continuous optimum

Additional Connections

$31^2 = 961 = 367 \times \phi^2$ (gap 0.019%), hence $G_{\text{Koch}} = \frac{367\sqrt{5}}{\phi^{10} \times 10^{11}}$. Mersenne property $31 = 2^5 - 1$ places 31 at binary-golden boundary.

Full precision value of G_{Koch} :

$$G_{\text{Koch}} = 6.6735467449616181237591997717904236079132174548715 \\ 1565557657672182332559409890102447913340675769273 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$$

Author: Boris V. Kochenov

Email: kochenovforwork@gmail.com

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